ITR2-1: Automatic Course Evaluation Extraction (AI-Optional & Robust Design)

As a student,
I want to upload my course outline (PDF or image) and automatically extract grading components and rules,
So that I can quickly set up my course evaluation without manual data entry.

Details

- Allow upload of PDF or image course outlines.
- Extract assessment components (tests, quizzes, assignments, exams).
- Extract weight percentages.
- Detect grading rules (Best X of Y, Drop Lowest, Mandatory Pass, Bonus Caps)
- Convert extracted data into structured grading format.
- Display extracted structure in editable preview.
- Require user confirmation before saving.
- Validate:
  - No negative weights
  - Total weight ≤ 100%
  - Logical rule consistency

AI-Optional Design

- Manual course setup remains available.
- System functions fully even if AI extraction fails.
- AI is used only for extraction, not for core grade calculations.

Acceptance Criteria

- User can upload a file successfully.
- System extracts assessments, weights, and supported rules.
- Extracted data is editable before saving.
- Validation prevents invalid grading structures.
- Core calculations operate independently of AI.

Priority: High
Estimated Cost: 5–6 days

ITR2-2: Persistent Multi-Course Planning System (DB + Multi-Course Support)

As a student,
I want my courses, grades, and what-if scenarios to be saved permanently,
So that I can track and plan my academic performance across sessions without losing data.
Details

Course Persistence

- Save extracted course structure (assessments, weights, grading rules).
- Load saved courses automatically on application restart.
- Allow editing and deleting saved courses.
- Ensure course data remains consistent after reload.

Score & Scenario Persistence

- Save entered grades per assessment.
- Allow updating and deleting stored scores.
- Save and label What-If scenarios.
- Reload scenarios without overwriting actual grades.
- Ensure recalculation occurs correctly after reload.

Multi-Course Planning

- Allow creation of multiple courses per student.
- Provide overview page listing:
  - Course name
  - Current grade
  - Target grade
  - Feasibility status (Achievable / Risky / Impossible)
- Allow switching between courses without data loss.
- Support cascading deletion of course-related data.
- Implement repository abstraction.
- Support both stub (in-memory) and PostgreSQL databases.
- Enable database switching via configuration.

Acceptance Criteria

- Students can create, edit, delete, and load multiple courses.
- Data persists after restarting the application.
- Scores and scenarios reload accurately.
- Switching between stub and PostgreSQL works without errors.
- Multi-course overview displays accurate summaries.
- Cascading deletion removes all related course data.
- Integration tests validate PostgreSQL persistence behaviour.

Priority: High
Estimated Cost: 4-5 days

ITR2-3: Universal GPA Converter (4.0, 9.0, & 10.0 Scales)

As a student, I want to convert my course percentages and YorkU grade points into various standardized GPA scales (4.0, 9.0, and 10.0), **so that** I can assess my academic standing for graduate school applications, internships, and internal university requirements.

Details

- **Multi-Scale Support:**
  - Implement logic for the **OMSAS 4.0 scale** (standard for medical/law applications).
  - Implement logic for the **YorkU 9.0 scale**.
  - Implement logic for a standard **10.0 scale** (common for international conversions).
- **Weighted GPA Calculation:**
  - Calculate cumulative GPA (cGPA) by weighting individual course grade points by their respective credit values (e.g., 3.0 vs 6.0 credits).
  - Provide a "Term GPA" vs. "Cumulative GPA" toggle.
- **Visual Data Representation:**
  - Create a summary table comparing the student's standing across all three scales simultaneously.
  - Highlight the conversion path (e.g., "82% $\rightarrow$ A $\rightarrow$ 3.8/4.0").
- **System Agnosticism:**
  - The converter must work for both "Actual" grades and "What-if" scenarios.
  - Ensure the conversion logic remains decoupled from the UI to allow for future scale additions (e.g., 7.0 or 12.0 scales).

Acceptance Criteria

- The system accurately converts a percentage to the correct Grade Point on 4.0, 9.0, and 10.0 scales.
- Users can input "Credit Weight" for courses to ensure the cGPA calculation is weighted correctly.
- The dashboard displays a side-by-side comparison of all three scales for the current semester.
- GPA calculations update dynamically when "What-if" scores are modified in the Strategy Dashboard.
- The system handles non-numeric grades (e.g., Pass/Fail or Withdrawn) without breaking the calculation engine.

Priority: High

Estimated Cost: 3–4 days

ITR2-4: Course Evaluation & Rule Modeling

As a student,
I want the system to evaluate my course performance using York University's grading rules and compute what I need to hit my targets,
So that I can understand my academic standing through letter grades, grade points, and realistic projections.

Details

- Map calculated percentages to the YorkU 9-point scale (A+ = 9, A = 8, etc.).
- Return structured results: Letter Grade, Grade Point Value, and Performance Descriptor (e.g., Exceptional, Excellent, Good).
- Calculate the exact average required on remaining coursework to achieve a user-defined target grade.
- Return transparent calculation fields (e.g., required points and required-average fraction).
- Categorize the target difficulty as Comfortable, Achievable, Challenging, Very Challenging, Not Possible, Already Achieved, or Complete.
- Ensure the evaluation engine is deterministic and read-only (no mutation of stored grade data).
- Stability: ensure consistent rounding at boundary thresholds (e.g., 79.5% vs 80.0%) and consistent results across repeated calls.
- Edge Cases:
    - 0% remaining weight (course complete).
    - Target already met (required average = 0%).
    - Required average exceeds 100% (Not Possible).

Implementation Notes

- Core grading calculations remain independent of AI extraction components to ensure 100% mathematical accuracy.
- Feature is compatible with both in-memory stub and persistent database implementations.
- Evaluation outputs include clear explanatory text for decision transparency.

Acceptance Criteria

- YorkU mapping is correct at boundaries (e.g., 90+ for A+, 80–89 for A).
- Required Remaining Average is mathematically correct for partially graded courses.
- Difficulty labels trigger correctly based on the computed required average.
- Running evaluation does not modify underlying course or stored grade objects.
- Inputs are validated (0–100) and "Not Possible" cases provide clear explanatory text.

Priority: High
Estimated Cost: 4–5 days

ITR2-5: Interactive Grade Modeling & Strategy Dashboard

As a student,
I want an interactive dashboard that models my best and worst-case grade scenarios and suggests specific study techniques,
So that I can strategically prioritize my efforts and trust the accuracy of the system's projections through transparent calculations.

Details
Interactive Planning Dashboard

- Create a central hub visualizing current progress and upcoming weights.
- Display overall GPA projections based on current and future data.
- Ensure projections update immediately when "what-if" scores are entered.

Grade Boundary Modeling

- Calculate and display "Worst-Case" scenarios (minimum scores to pass/hit target).
- Calculate and display "Best-Case" scenarios (maximum possible reachable grade).
- Handle calculations accurately even if total course weightings do not sum to 100%.

Calculation Transparency

- Implement a "Show Calculations" toggle/expandable button for all estimates.
- Display the underlying mathematical logic (e.g., weighted averages).
- Allow students to manually verify the system's projection logic.

Smart Learning Strategies

- Integrate a recommendation engine for specific learning techniques.
- Suggest the 80/20 Rule (Pareto Principle) for high-weight, high-density topics.
- Provide context-aware suggestions (Active Recall, Spaced Repetition, Feynman Technique) based on assessment type and time remaining.

Acceptance Criteria

- The dashboard displays a clear "Max/Min" boundary for the final course grade.
- An "Expand/Show Math" button successfully reveals the formulas used.
- The system suggests at least one specific learning technique for each major deadline.
- The UI handles "what-if" scenarios without overwriting actual saved grades.
- Calculations remain accurate during incomplete syllabi (weights < 100%).

Priority: High
Estimated Cost: 5–6 days

ITR2-6: Smart Deadline Management (OCR + Manual Entry + Calendar Sync)

As a student**,**
I want my assignment and test deadlines to be automatically extracted from my course outline, but also be able to manually add or edit them,
so that I have full control over my deadlines and never miss anything due to OCR errors or missing information

Details

- Use OCR to extract assessment names, due dates, and times (if available) from uploaded course outlines.
- Display all detected deadlines in an editable list view.
- Allow users to:
  - Edit any deadline (name, date, time, type, notes).
  - Delete incorrect entries.
  - Add new deadlines manually at any time.
- If OCR detects no deadlines, show a clear empty state with an "Add Deadline" button.
- Include a review & confirm step before exporting deadlines to Google Calendar.
- Allow users to export selected or all deadlines to Google Calendar using the Google Calendar API.
- Exported events should include:
  - Course name + assessment title.
  - Due date/time.
  - Automatic reminder 1 week before the due date.
  - The minimum grade needed (if calculated) in the event description.
- Show a countdown display on the dashboard for upcoming deadlines (e.g., "3 days left until your final exam, where you need a minimum grade of 63 to meet your target goal").
- Mark deadlines with a badge (e.g., "From Outline" or "Manual") for clarity.

Acceptance Criteria

- OCR-detected deadlines appear in a structured, editable list.
- Users can add, edit, or delete any deadline (manual or OCR-generated).
- Manual deadlines function exactly the same as OCR-detected ones (countdown, reminders, export).
- Countdown updates dynamically based on the current date.
- Exported events appear correctly in the user's selected Google Calendar.
- A reminder 7 days before each due date is created automatically.
- If a minimum grade requirement exists, it is included in the calendar event details.
- The system prevents duplicate exports of the same deadline.

Priority: Medium
Estimated Cost: 4–5 days